

CASIMIR-LIFSHITZ EFFECT IN WATER CONDENSATION

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We consider the Lifshitz theory for a three layered concentric sphere geometry where the center particle can be sub-micron sized and has the dielectric properties of cellulose, the middle layer has the dielectric properties of either liquid water or ice and the infinitely extending outer layer has the dielectric properties of water vapor ($\epsilon = \epsilon_0$). This serves as a toy model for studying the Lifshitz effect on seeding particles in clouds. We show that the Lifshitz energy is minimized for a finitely thick water layer, which we refer to as the minimizing thickness. By analyzing changes to the saturation vapor pressure, we find the Lifshitz effect strongly promotes water condensation below the energy minimizing layer-thickness and slightly inhibiting it above.

Casimir-Lifshitz Theory

The Casimir-Lifshitz effect accounts for the quantum and thermal fluctuation induced interactions between two or more macroscopic bodies. The dielectric functions (ϵ_i) and geometry of each material determines the interaction energy in thermal equilibrium at a given temperature.

1 Spherical Case [1]

The interaction energy without divergent terms for the concentric sphere Lifshitz effect with cellulose in the center sphere $\epsilon_{\text{cellulose}} = \epsilon_1$, covered by water $\epsilon_{\text{liquid water/ice}} = \epsilon_3$ in a vapor environment $\epsilon_{\text{vapor}} = \epsilon_2$ is given by:

$$F^s(b, a) = \sum_{n=0}^{\infty} \sum_{l=1}^{\infty} \sum_{\sigma} \frac{2\nu}{\beta} \ln[1 - \tilde{r}_{\sigma}^{31}(a) \tilde{r}_{\sigma}^{32}(b)], \quad (1)$$

where $\sigma = TE, TM$, represent the transverse electric and magnetic modes which is summed over, the primed sum over the Masturbara modes, $\omega_n = \frac{2\pi n k_B T}{h}$ means that the n=0 term contains a factor of 1/2. The inner radius (a) and the outer radius (b) are represented in the picture to the right and $\nu = l + \frac{1}{2}$.

$$\tilde{r}_{TE}^{31}(a) = \frac{\zeta_3 i_l(\zeta_1 a) \bar{i}_l(\zeta_3 a) - \zeta_1 \bar{i}_l(\zeta_1 a) i_l(\zeta_3 a)}{\zeta_3 i_l(\zeta_1 a) \bar{k}_l(\zeta_3 a) - \zeta_1 \bar{i}_l(\zeta_1 a) k_l(\zeta_3 a)} \quad \tilde{r}_{TM}^{31}(a) = \frac{\zeta_1 i_l(\zeta_1 a) \bar{i}_l(\zeta_3 a) - \zeta_3 \bar{i}_l(\zeta_1 a) i_l(\zeta_3 a)}{\zeta_1 i_l(\zeta_1 a) \bar{k}_l(\zeta_3 a) - \zeta_3 \bar{i}_l(\zeta_1 a) k_l(\zeta_3 a)} \quad (2)$$

$$\tilde{r}_{TE}^{32}(b) = \frac{\zeta_2 k_l(\zeta_3 b) \bar{k}_l(\zeta_2 b) - \zeta_3 \bar{k}_l(\zeta_3 b) k_l(\zeta_2 b)}{\zeta_2 i_l(\zeta_3 b) \bar{k}_l(\zeta_2 b) - \zeta_3 \bar{i}_l(\zeta_3 b) k_l(\zeta_2 b)} \quad \tilde{r}_{TM}^{32}(b) = \frac{\zeta_3 k_l(\zeta_3 b) \bar{k}_l(\zeta_2 b) - \zeta_2 \bar{k}_l(\zeta_3 b) k_l(\zeta_2 b)}{\zeta_3 i_l(\zeta_3 b) \bar{k}_l(\zeta_2 b) - \zeta_2 \bar{i}_l(\zeta_3 b) k_l(\zeta_2 b)} \quad (3)$$

The reflection coefficients, r_{σ}^{ij} , are in terms of modified spherical Bessel functions i_l and k_l , with their generalized derivatives being: $\bar{k}_l(t) = (\frac{1}{t} + \frac{\partial}{\partial t})k_l(t)$ and $\bar{i}_l(t) = (\frac{1}{t} + \frac{\partial}{\partial t})i_l(t)$ and $\zeta_i = \frac{\omega_n}{hc} \sqrt{\epsilon_i(i\omega_n)}$.

2 Planar Cases: Three and Five Layer [2]

As $a \rightarrow \infty$, we arrive at the three layered planar interaction energy density (keeping the ϵ_i labels the same):

$$F^{3p}(d) = \frac{k_B T}{2\pi} \sum_{m=0}^{\infty} \int_0^{\infty} dq q \sum_{\sigma} \ln(1 - r_{\sigma}^{31} r_{\sigma}^{32} e^{-2\kappa_2 d}), \quad (4)$$

d is the distance indicated in the above figure. Now the reflection coefficients ($\kappa_i = \sqrt{q^2 + \frac{\epsilon_i \omega_n^2}{c^2}}$, $\epsilon_i = \epsilon_i(i\omega_n)$)

$$r_{TE}^{ij} = \frac{\kappa_i - \kappa_j}{\kappa_i + \kappa_j}; \quad r_{TM}^{ij} = \frac{\epsilon_j \kappa_i - \epsilon_i \kappa_j}{\epsilon_j \kappa_i + \epsilon_i \kappa_j}. \quad (5)$$

Notice r_{σ}^{ij} changes sign when the dielectric function of adjoining materials cross (i.e. $\epsilon_1 = \epsilon_3$). From the pressure expression below this, along with the distance, determines if or when the pressure changes signs.

$$\frac{\partial F}{\partial d} = \frac{k_B T}{2\pi} \sum_{m=0}^{\infty} \int_0^{\infty} dq q \sum_{\sigma} \frac{2\kappa_2 r_{\sigma}^{31} r_{\sigma}^{32} e^{-2\kappa_2 d}}{(1 - r_{\sigma}^{31} r_{\sigma}^{32} e^{-2\kappa_2 d})} \quad (6)$$

Now, consider a 1D slice (line) passing the sphere's origin which contains five dielectric segments. The outer two segments are infinitely extending water vapor (ϵ_2). The center is cellulose (ϵ_1) of length $t = 2a$, sandwiched between two water segments of length d (ϵ_3). Extending each segment to an infinite slab with same thickness, the geometry becomes a five layer planar one and the energy expression is given by:

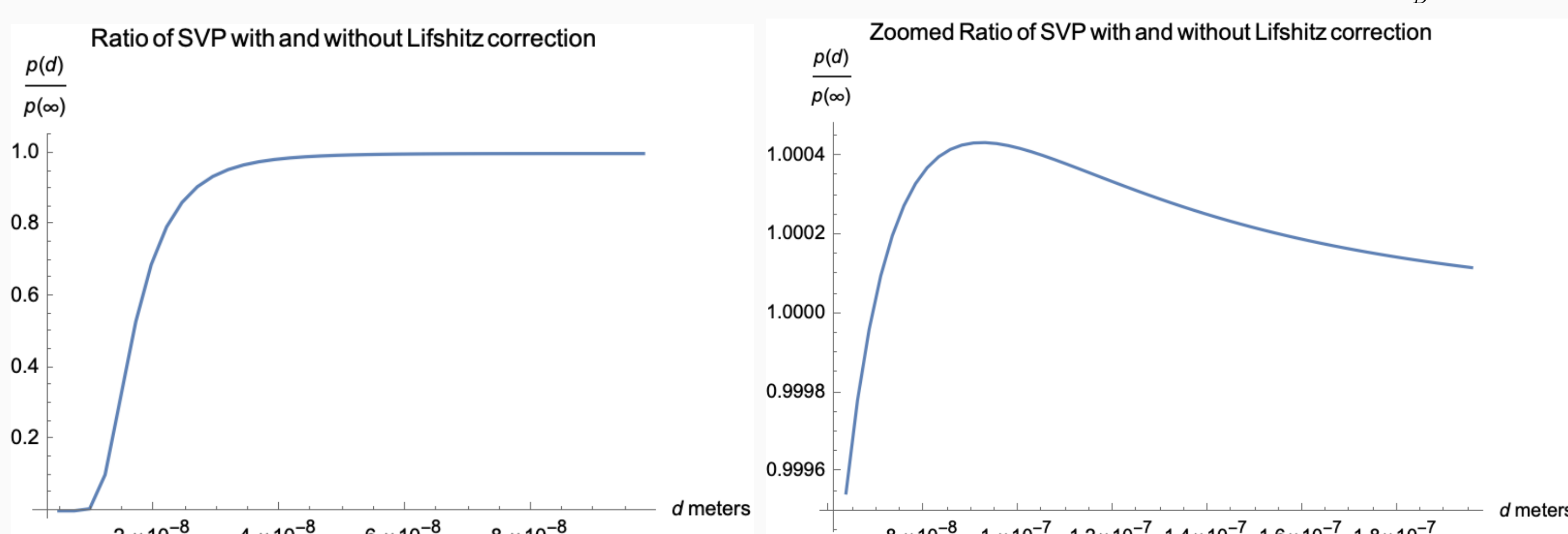
$$F^5(d) = \frac{k_B T}{2\pi} \sum_{m=0}^{\infty} \int_0^{\infty} dk k \sum_{\sigma} \ln[1 + 2r_{\sigma}^{23} r_{\sigma}^{31} e^{-2\kappa_3 d} (1 - e^{-2\kappa_1 t}) - (r_{\sigma}^{31})^2 e^{-2\kappa_1 t} + (r_{\sigma}^{23} r_{\sigma}^{31})^2 e^{-4\kappa_3 d} - (r_{\sigma}^{23})^2 e^{-4\kappa_3 d} e^{-2\kappa_1 t}]$$

Disjoining Pressure and Saturation Vapor Pressure

We conclude that the Casimir-Lifshitz effect strongly promotes water condensation when water layer is less than the minimizing thickness and weakly inhibits water condensation for larger layers by respectively lowering/raising the saturation vapor pressure (SVP) relative to the value for an infinitely thick water layer $p_s(\infty)$. The Gibbs free energy after a thin water layer has formed is $dG = (\mu_v - \mu_w) dN_v + \sigma dA + \frac{\partial F^s}{\partial b} db$, and the SVP is:

$$p_s(a, b) = p_s(\infty) \text{Exp}\left(\frac{v_{mw}}{b^2 k_B T} (2\sigma b + \frac{\partial F^s(b, a)}{\partial b})\right) \quad (7)$$

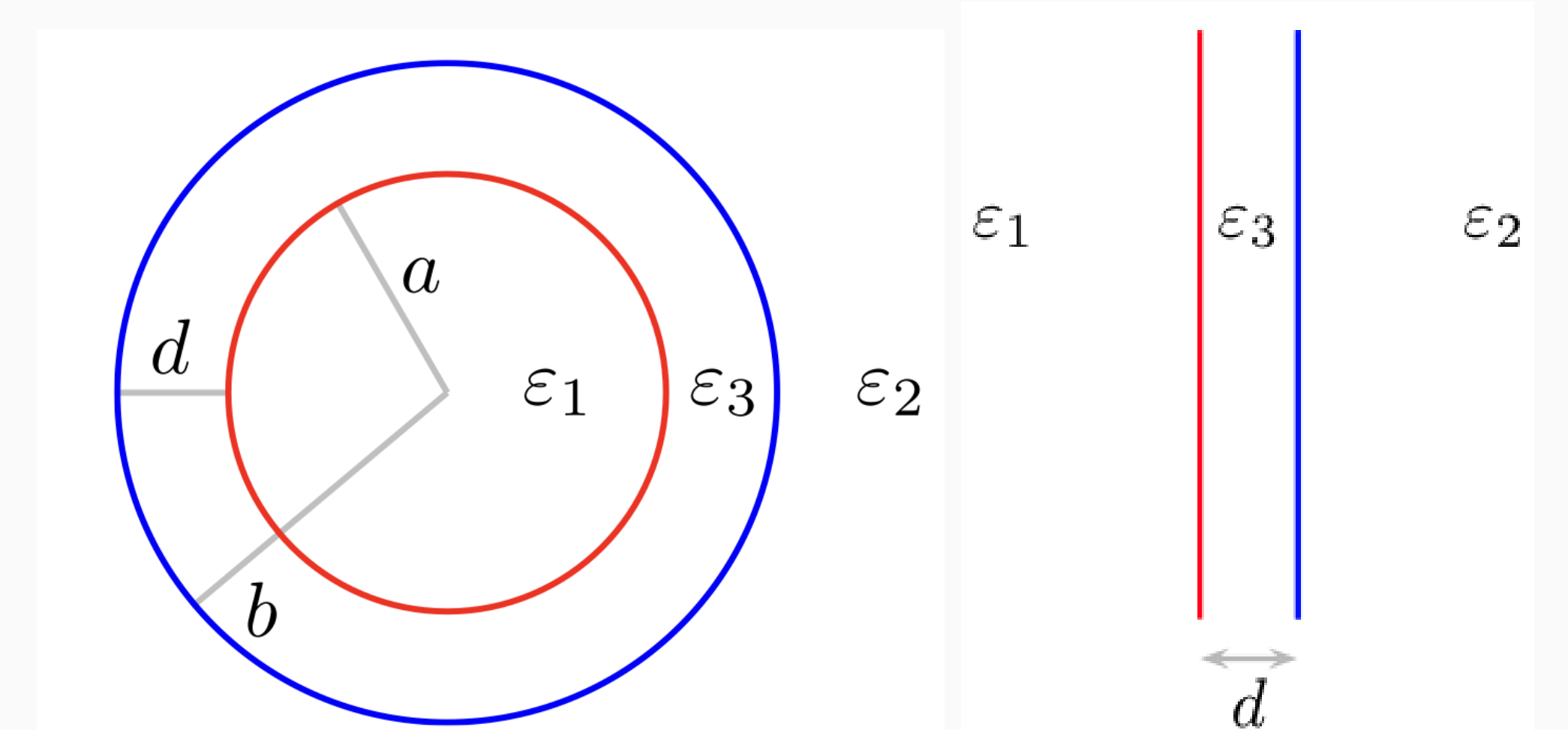
The blue term is the surface energy contribution for drop of water of radius b, which is negligible for $b = d+a > 1$ nm, and v_{mw} is the volume of one water molecule. In the plots, we illustrate the ratio of SVPs ($p_s(d)/p_s(\infty)$) in three layered planar geometry for which the expression simplifies to $p_s(d) = p_s(\infty) \text{Exp}(\frac{v_{mw}}{k_B T} \frac{\partial F^{3p}(d)}{\partial d})$



$\frac{p_s(d)}{p_s(\infty)} = 1$ only at the minimizing thickness. With $p_s(\infty) = 0.02$ atm, the maximum the Lifshitz term can inhibit condensation by is $\approx 8 \times 10^{-6}$ atm but can drastically lower the SVP promoting condensation.

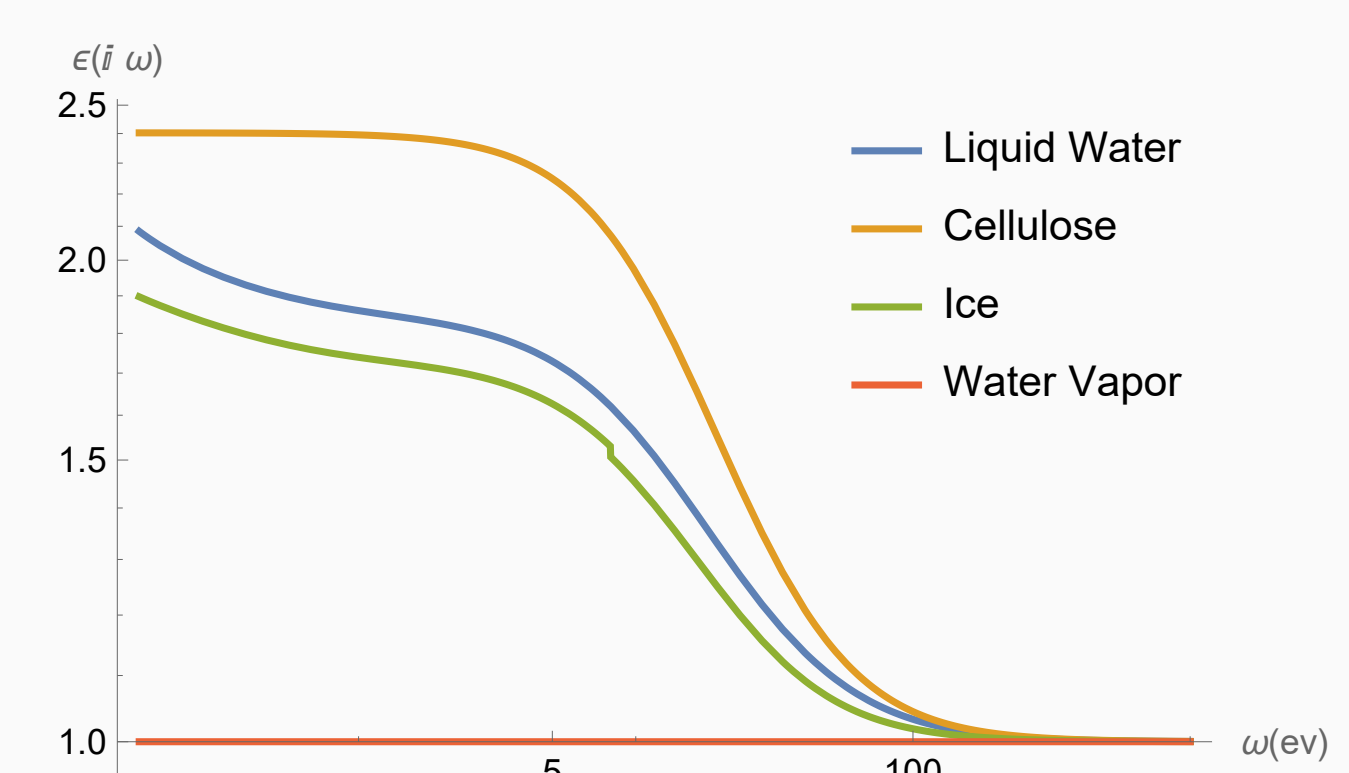
Geometric Figures [1]

Left: Three layered concentric sphere **Right:** Three layered planar

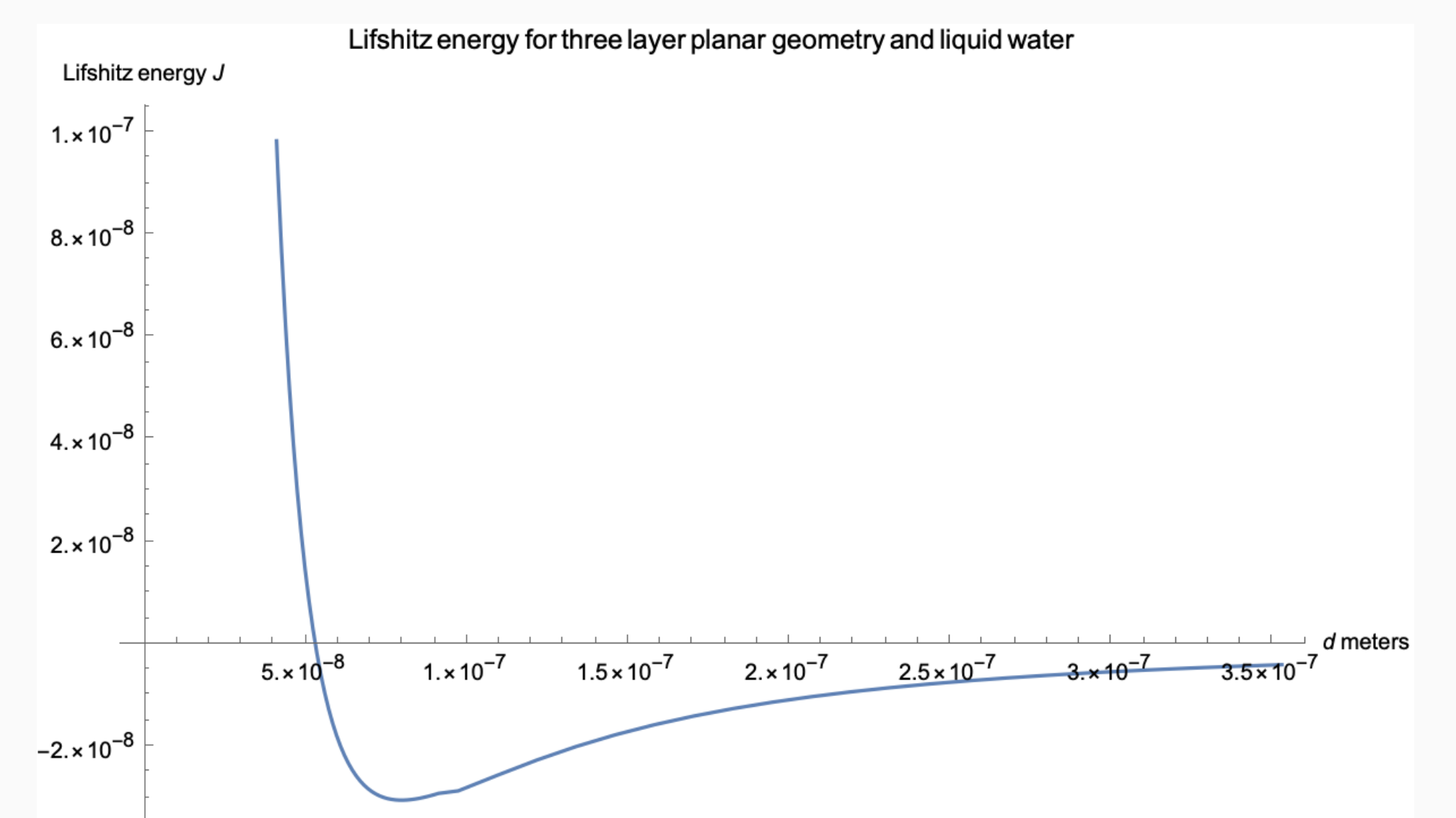


Data

The zero frequency for the dielectric functions $\epsilon_{\text{cellulose}}(0) = 2.602$, $\epsilon_{\text{liquid water}}(0) = 78.74$, and $\epsilon_{\text{ice}}(0) = 91.5$, are not included in the graph but indicate a long-distance crossing between water/ice and cellulose. $\epsilon(i\omega)$ can be related to $\epsilon_1 = \text{Re}(\epsilon)$ and $\epsilon_2 = \text{Im}(\epsilon)$ via Kramers-Kronig.



Lifshitz energy with liquid water is plotted in the planar limit:



The table below shows that as the cellulose particle shrinks, the liquid water/ice thickness which minimizes the Lifshitz energy increases.

Minimizing thickness of ϵ_3 for cellulose particle		
Nucleus Radius	$\epsilon_3 = \epsilon_{\text{water}}$	$\epsilon_3 = \epsilon_{\text{ice}}$
∞ (plane)	71 nm	80.7 nm
1 μm	72.2 nm	81.9 nm
100 nm	88.1 nm	103.9 nm
10 nm	128.5 nm	149.1 nm

Unlike the concentric sphere results, as the width of cellulose slab decreases in the five layer planar geometry so does the water's minimizing thickness, matching the three layer result for at least $t > 0.2 \mu\text{m}$.

Minimizing thickness of ϵ_3 for cellulose slab		
Slab width (t)	$\epsilon_3 = \epsilon_{\text{water}}$	$\epsilon_3 = \epsilon_{\text{ice}}$
200nm	71 nm	80.7 nm
20 nm	55 nm	59.9 nm
2 nm	18 nm	20.8 nm
.2 nm	1.7 nm	2.5 nm

Acknowledgements and References

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[1]Parashar, Prachi, et al. "Role of zero point energy in promoting ice formation in a spherical drop of water." PRR. h 1.3 (2019): 033210.

[2]Li, Y., et al. "Origin of anomalously stabilizing ice layers on methane gas hydrates near rock surface." PCCP. 25.9 (2023): 6636-6652.